Set7 A. 4.25 4.31 4.33 B 5.2.5.10.5.19

4.25 解 $\gamma_1 = \frac{1}{\sqrt{1-(\frac{v_1}{c})^2}}$ $\gamma_1 = \frac{5}{4}$ $(m_0 = 9.1\times10^{-31}(kg))$

$m_1 = \gamma_1 m_0$

$E_{K1} = m_1c^2 - m_0c^2$ $\Rightarrow E_{K1} = \frac{1}{4}m_0c^2$ $E_{K1} = 2.05\times10^{-14}$ J

$\gamma_2 = \frac{1}{\sqrt{1-(\frac{v_2}{c})^2}}$ $\gamma_2 = \frac{5}{3}$

$m_2 = \gamma_2 m_0$

$E_{K2} = m_2c^2 - m_1c^2$ $\Rightarrow E_{K2} = \frac{5}{12}m_0c^2$ $E_{K2} = 3.41\times10^{-13}$

4.31 解 $(E_v + mc^2)$

$$\begin{cases} m_Nc^2 = E_M + E_v \\ E_v = P_vc \\ E_M^2 = P_M^2c^2 + m_M^2c^4 \\ P_M + P_v = 0 \end{cases} \Rightarrow \begin{cases} E_M = \frac{m_N^2 + m_M^2}{2m_N}c^2 \\ E_v = \frac{m_N^2 - m_M^2}{2m_N}c \end{cases}$$

$E_{KM} = E_M - m_Mc^2 = \frac{(m_N - m_M)^2}{2m_N}$

4.33 解 $\gamma_A = \frac{1}{\sqrt{1-(\frac{v_A}{c})^2}}$ $\gamma_A = \frac{5}{4}$

$m_1 = \gamma_A m_0$

$E_{K1} = m_1c^2 - m_0c^2$ $E_{K1} = \frac{1}{4}m_0c^2$

$P_1 = m_1v_A$ $P_1 = \frac{3}{4}m_0c$

$v_A' = \frac{v_A + v}{1 + \frac{vv_A}{c^2}}$ $v_A' = \frac{11}{13}c$ $\gamma_A = 1.876$

$E_{K2} = m_2c^2 - m_0c^2$ $E_{K2} = 0.876m_0c^2$

$P_2 = m_2v_A'$ $P_2 = 1.58m_0c$

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$x = A\cos(\omega t + \varphi)$

5.2 解: (1) $\omega t_a + \varphi = 0$

$\omega t_b + \varphi = \frac{2\pi}{3}$

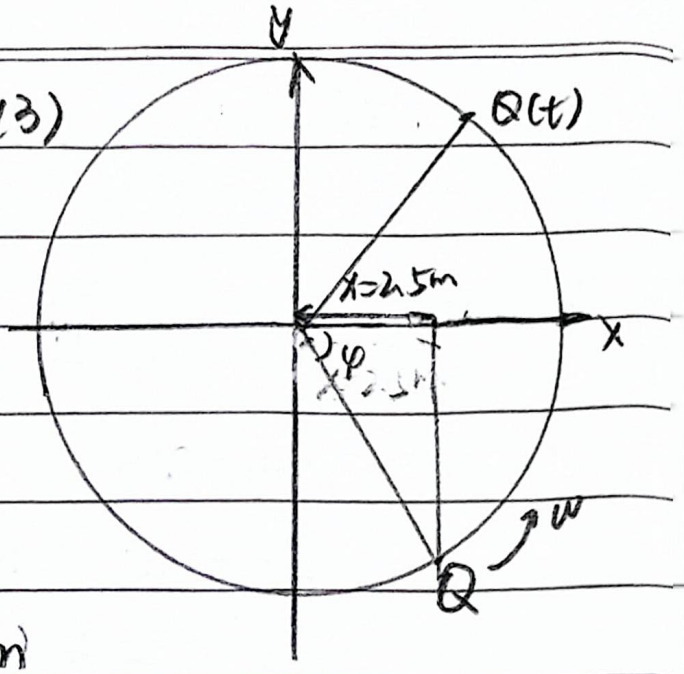
$\omega t_c + \varphi = \frac{\pi}{2}$

$\omega t_d + \varphi = \frac{2}{3}\pi$

$\omega t_e + \varphi = \frac{4}{3}\pi$

(2) $x = 0.05\cos(\frac{5}{6}\pi t - \frac{\pi}{3})$ (m)

(3)



5.10 解: (1) $x = 0.12\cos(\frac{4}{3}\pi t)$ (m)

$t = 0.5$ s 时, $x = -0.12$ m

(2) $v = \frac{dx}{dt} = -\frac{8}{5}\pi\sin\frac{4}{3}\pi t$ m/s

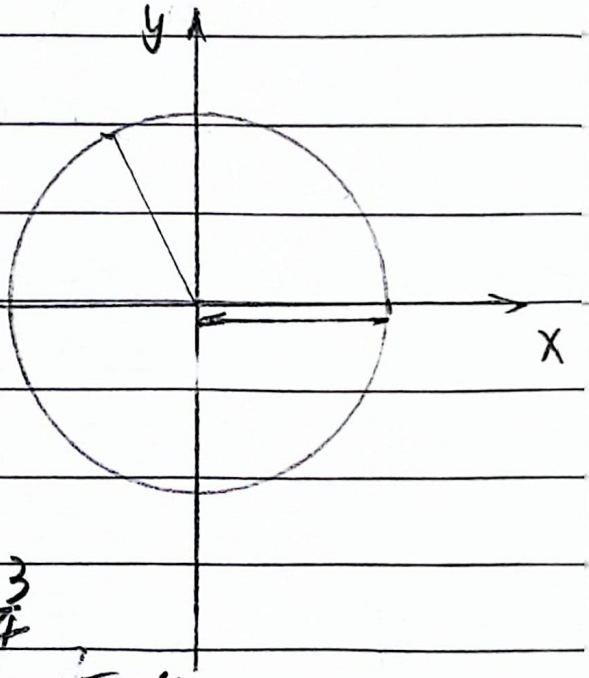
$a = \frac{dv}{dt} = -\frac{32}{75}\pi^2\cos\frac{4}{3}\pi t$ m/s²

$F = ma|_{t=0.5s} = \frac{32}{75}\pi^2 \times 10^{-2}$ N

方向向 x 轴正向

(3) $x = -0.18$ m $\Rightarrow t = \frac{3}{4\pi}\arccos(-\frac{3}{4})$

(4) $v^2 = \omega^2(A^2 - x^2) \Rightarrow v = 0.665$ m/s



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5.19 $K = \frac{mg}{\Delta x}$

$\omega = \sqrt{\frac{K}{m}}$ $\omega = 5\pi\ 10\ rad/s$

$A^2 = \frac{v^2}{\omega^2} + x^2$ $\Rightarrow A = 0.033m$

$\Rightarrow x = 0.033 \cos(15.8t + \frac{2}{5}\pi)$ m

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